40 A student investigates four different radioactive isotopes. The student places a detector near each radioactive material.

The background count rate is 36 counts per minute throughout the investigation.

The table shows the detector readings at the start and after 8 hours.

Which isotope has a half-life of 4 hours?

	count rate at the start counts per minute	count rate after 8 hours counts per minute
A	150	36
B	212	53
C	260	92
D	356	80